

1. Statement $(P \Rightarrow Q) \wedge (R \Rightarrow Q)$ is logically equivalent to

[2023 (06 Apr Shift 1)]

(1) $(P \Rightarrow R) \vee (Q \Rightarrow R)$

(2) $(P \wedge R) \Rightarrow Q$

(3) $(P \Rightarrow R) \wedge (Q \Rightarrow R)$

(4) $(P \vee R) \Rightarrow Q$

2. Among the statements

(S1): $(p \Rightarrow q) \vee ((\neg p) \wedge q)$ is a tautology

(S2): $(q \Rightarrow p) \Rightarrow ((\neg p) \wedge q)$ is a contradiction

[2023 (06 Apr Shift 2)]

(1) Neither (S1) and (S2) is True

(2) Both (S1) and (S2) are True

(3) Only (S2) is True

(4) Only (S1) is True

3. Negation of $(p \rightarrow q) \rightarrow (q \rightarrow p)$ is

[2023 (08 Apr Shift 1)]

(1) $(p \neg) \vee p$

(2) $q \wedge (\neg p)$

(3) $(\neg q) \wedge p$

(4) $p \vee (\neg q)$

4. The negation of $(p \wedge (\neg q)) \vee (\neg p)$ is equivalent to

[2023 (08 Apr Shift 2)]

(1) $p \wedge (\neg q)$

(2) $p \wedge q$

(3) $p \vee (q \vee (\neg p))$

(4) $p \wedge (q \wedge (\neg p))$

5. The negation of the statement $(p \vee q) \wedge (q \vee (\neg r))$ is

[2023 (10 Apr Shift 1)]

(1) $(p \vee r) \wedge (\neg q)$

(2) $((\neg p) \vee r) \wedge (\neg q)$

(3) $((\neg p) \vee (\neg q)) \vee (\neg r)$

(4) $((\neg p) \vee (\neg q)) \wedge (\neg r)$

6. The statement $\neg[p \vee (\neg(p \wedge q))]$ is equivalent to

[2023 (10 Apr Shift 2)]

(1) $\neg(p \wedge q)$

(2) $(p \wedge q) \wedge (\neg p)$

(3) $(\neg(p \wedge q)) \wedge q$

(4) $\neg(p \vee q)$

7. The number of ordered triplets of the truth values of p, q and r such that the truth value of the statement $(p \vee q) \wedge (p \vee r) \Rightarrow (q \vee r)$ is True, is equal to

[2023 (11 Apr Shift 1)]

8. The converse of $((\neg p) \wedge q) \Rightarrow r$ is

[2023 (11 Apr Shift 2)]

(1) $((\neg p) \vee q) \Rightarrow r$

(2) $(\neg r) \Rightarrow p \wedge q$

(3) $(\neg r) \Rightarrow ((\neg p) \wedge q)$

(4) $(p \vee (\neg q)) \Rightarrow (\neg r)$

9. Among the two statements

(S₁): $(p \Rightarrow q) \wedge (p \wedge (\neg q))$ is a contradiction and (S₂): $(p \wedge q) \vee ((\neg p) \wedge q) \vee (p \wedge (\neg q)) \vee ((\neg p) \wedge (\neg q))$ is a tautology

[2023 (12 Apr Shift 1)]

(1) only (S₂) is true

(2) only (S₁) is true

(3) both are false

(4) both are true

10. The negation of the statement $((A \wedge (B \vee C)) \Rightarrow (A \vee B)) \Rightarrow A$ is _____
[2023 (13 Apr Shift 1)]
- (1) equivalent to $\sim C$
 - (2) equivalent to $B \vee \sim C$
 - (3) a fallacy
 - (4) equivalent to $\sim A$
11. The statement $(p \wedge (\sim q)) \vee ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q))$ is equivalent to _____
[2023 (13 Apr Shift 2)]
- (1) $(\sim p) \vee q$
 - (2) $(\sim p) \vee (\sim q)$
 - (3) $p \vee (\sim q)$
 - (4) $p \vee q$
12. Negation of $p \wedge (q \wedge \sim(p \wedge q))$ is _____
[2023 (15 Apr Shift 1)]
- (1) $(\sim(p \wedge q)) \vee p$
 - (2) $p \vee q$
 - (3) $\sim(p \vee q)$
 - (4) $(\sim(p \wedge q)) \wedge q$

ANSWER KEYS

1. (4) 2. (1) 3. (2) 4. (2) 5. (2) 6. (2) 7. (7) 8. (4)
9. (4) 10. (4) 11. (2) 12. (1)

1. (4)

Given that $(P \Rightarrow Q) \wedge (R \Rightarrow Q)$

We know that $P \Rightarrow Q \equiv \neg P \vee Q$

$\Rightarrow (\neg P \vee Q) \wedge (\neg R \vee Q)$

$\Rightarrow (\neg P \wedge \neg R) \vee Q$

$\Rightarrow \neg(P \vee R) \vee Q$

$\Rightarrow (P \vee R) \Rightarrow Q$

Hence this is the correct option.

2. (1)

Plotting the Truth table of the given expression we get,

p	q	$\neg p$	$\neg p \wedge q$	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \vee (\neg p \wedge q)$	$(q \Rightarrow p) \vee (\neg p \wedge q)$
T	T	F	F	T	T	T	T
T	F	F	F	F	T	F	T
F	T	T	T	T	F	T	T
F	F	T	F	T	T	T	T

Hence, $(q \Rightarrow p) \vee (\neg p \wedge q)$ is a tautology, and $(p \Rightarrow q) \vee (\neg p \wedge q)$ is neither tautology and contradiction,

So both statements are false.

3. (2)

We know that, $A \rightarrow B \equiv \neg A \vee B$

Now using the above formula we will solve,

$(p \rightarrow q) \rightarrow (q \rightarrow p)$

$\equiv \neg(\neg p \vee q) \vee (\neg q \vee p)$

$\equiv (p \wedge \neg q) \vee (\neg q \vee p)$

$\equiv \neg q \vee p$

Now finding the negation of $(p \vee \neg q)$ we get,

$\neg(p \vee \neg q)$

$\equiv \neg p \wedge q$

4. (2)

The given statement is $(p \wedge (\neg q)) \vee (\neg p)$

Let us take $(A \cap B^c) \cup (A^c)$.

Now let us apply distributive law to the above statement.

$\equiv (A \cup A^c) \cap (B^c \cup A^c)$

$\equiv (U) \cap (B^c \cup A^c)$

$\equiv (B^c \cup A^c)$

Now the given statement $(p \wedge (\neg q)) \vee (\neg p)$ will be equivalent to $\neg q \vee \neg p$ or $\neg p \vee \neg q$.

Hence the given statement is equivalent to $\neg p \vee \neg q$.

Now negation of given statement by demorgan's law will be,

$\neg(\neg p \vee \neg q) \equiv p \wedge q$

5. (2)

Given,

$$(p \vee q) \wedge (q \vee (\neg r))$$

Now negation of above expression will be,

$$\neg((p \vee q) \wedge (q \vee (\neg r)))$$

$$= \neg(p \vee q) \vee \neg(q \vee (\neg r)) \quad \{\text{as } \neg(A \wedge B) \equiv \neg A \vee \neg B\}$$

$$= (\neg p \wedge \neg q) \vee (\neg q \wedge r)$$

$$= (\neg p \wedge \neg q) \vee (\neg q) \wedge (\neg p \wedge \neg q) \vee r$$

$$= \neg q \wedge [(\neg p \vee r) \wedge (\neg q \vee r)]$$

$$= (\neg q \wedge (\neg p \vee r)) \wedge (\neg q \wedge (\neg q \vee r))$$

$$= (\neg q \wedge (\neg p \vee r)) \wedge \neg q$$

$$= ((\neg p) \vee r) \wedge (\neg q)$$

6. (2)

Given,

$$\neg[p \vee (\neg(p \wedge q))]$$

Now we know that,

$$\text{By demorgan law, } \neg(A \vee B) \equiv \neg A \wedge \neg B$$

Now using above formula we get,

$$\neg[p \vee (\neg(p \wedge q))]$$

$$\equiv \neg p \wedge (p \wedge q) \quad [\text{as } \neg(\neg A) \equiv A]$$

$$\equiv (p \wedge q) \wedge (\neg p)$$

7. (7)

Given,

$$\text{Expression } (p \vee q) \wedge (p \vee r) \Rightarrow (q \vee r)$$

Now by truth table we get,

p	q	r	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$	$q \vee r$	$(p \vee q) \wedge (p \vee r) \Rightarrow (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	T	T	T	T	T
T	F	F	T	T	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	F	T	F	T	T
F	F	F	F	F	F	F	T

Now from above truth table we can say that there are total 7 triplet are possible.

8. (4)

We know that,

$$\text{Converse of } A \Rightarrow B \text{ is } B \Rightarrow A$$

So, using the above formula we get,

$$\text{Converse of } (\neg p \wedge q) \Rightarrow r$$

$$\equiv r \Rightarrow (\neg p \wedge q)$$

$$\equiv (\neg r) \vee (\neg p \wedge q)$$

$$\equiv (\neg p \wedge q) \vee (\neg r)$$

$$\equiv \neg(p \vee \neg q) \vee (\neg r) \quad \{\text{as } \neg(A \vee B) \equiv \neg A \wedge \neg B\}$$

$$\equiv (p \vee (\neg q)) \Rightarrow (\neg r)$$

9. (4)

Given,

$$S_1 : (p \Rightarrow q) \wedge (p \wedge \sim q)$$

$$\equiv (\sim p \vee q) \wedge (p \wedge \sim q) \text{ \{as } A \Rightarrow B \equiv \sim A \vee B \}$$

$$\equiv (\sim p \wedge (p \wedge \sim q)) \vee (q \wedge (p \wedge \sim q))$$

$$\equiv (f \wedge \sim q) \vee (f \wedge p)$$

$$\equiv f \vee f \equiv f$$

$$(S_2) : (p \wedge q) \vee ((\sim p) \wedge q) \vee (p \wedge (\sim q)) \vee ((\sim p) \wedge (\sim q))$$

$$\equiv (p \wedge q) \vee (\sim p \wedge q) \vee (p \wedge \sim q) \vee (\sim p \wedge \sim q)$$

$$\equiv ((p \vee \sim p) \wedge q) \vee ((p \vee \sim p) \wedge \sim q)$$

$$\equiv (t \wedge q) \vee (t \wedge \sim q) \equiv q \vee \sim q \equiv t$$

Hence, both statement are true.

10. (4)

Given,

$$((A \wedge (B \vee C)) \Rightarrow (A \vee B)) \Rightarrow A$$

We know that, $X \Rightarrow Y \equiv \sim X \vee Y$,

Now using the above formula we get,

$$\equiv (\sim (A \wedge (B \vee C)) \vee (A \vee B)) \Rightarrow A$$

$$\equiv \sim (\sim (A \wedge (B \vee C)) \vee (A \vee B)) \vee A$$

$$\equiv (A \wedge (B \vee C)) \wedge \sim (A \vee B) \vee A$$

Now to solve $(A \wedge (B \vee C)) \wedge \sim (A \vee B)$ let us assume a universal set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, set $A = \{1, 2, 3\}$, set $B = \{3, 4, 5\}$ and set $C = \{5, 6, 7\}$

So, $A \wedge (B \vee C)$ will be $\{3\}$ and $\sim (A \vee B)$ will be $\{6, 7, 8\}$,

So, $(A \wedge (B \vee C)) \wedge \sim (A \vee B) = \emptyset$

$$\equiv F \vee A$$

$$\equiv A$$

$\therefore$ Negation of statement = $\sim A$

11. (2)

Given,

$$(p \wedge (\sim q)) \vee ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q))$$

$$= ((\sim p) \wedge q) \vee ((\sim p) \wedge (\sim q)) \vee (p \wedge (\sim q))$$

$$= (\sim p \wedge (q \vee \sim q)) \vee (p \wedge \sim q)$$

We know that $\sim A \vee A = T$, where T is tautology,

$$= (\sim p \wedge T) \vee (p \wedge \sim q)$$

$$= \sim p \vee (p \wedge \sim q)$$

$$= (\sim p \vee p) \wedge (\sim p \vee \sim q)$$

$$= T \wedge (\sim p \vee \sim q)$$

$$= \sim p \vee \sim q$$

12. (1)

Given,

Expression $p \wedge (q \wedge \sim(p \wedge q))$

So, negation of the above expression will be,

$$\sim[p \wedge (q \wedge \sim(p \wedge q))]$$

Now using demorgan's law $\sim(A \wedge B) \equiv \sim A \vee \sim B$ & $\sim(A \vee B) \equiv \sim A \wedge \sim B$ we get,

$$\sim[p \wedge (q \wedge \sim(p \wedge q))]$$

$$\equiv \sim[p \wedge (q \wedge \sim(p \vee \sim q))]$$

$$\equiv \sim[p \wedge ((q \wedge \sim p) \vee (q \wedge \sim q))]$$

$$\equiv \sim[p \wedge ((q \wedge \sim p) \vee F)]$$

$$\equiv \sim[p \wedge (q \wedge \sim p)]$$

$$\equiv \sim p \vee \sim(q \wedge \sim p)$$

$$\equiv \sim p \vee (\sim q \vee p)$$

$$\equiv (\sim p \vee \sim q) \vee p$$

$$\equiv \sim(p \wedge q) \vee p$$